

Oxygenation versus tissue composition in positronium-lifetime PET: an identifiability limit for hypoxia imaging

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Abstract

Purpose. Positronium lifetime imaging with long-axial-field-of-view PET/CT has been proposed as a biomarker of tissue oxygenation. The same lifetime, however, also depends on tissue composition. We quantified whether oxygenation and composition can be separated from positronium observables and what bias arises if composition is ignored.

Methods. A rate-additivity model mapped pO_2 and lipid fraction to the ortho-positronium lifetime τ_3 and intensity I_3 . We derived the apparent- pO_2 bias of a water-calibrated reader, analysed structural identifiability by Fisher information, computed Cramér-Rao bounds (CRLBs) for (pO_2, f) from (τ_3, I_3) using a Poisson lifetime-spectrum model with nuisance profiling, and translated the bounds into counts and region-of-interest volumes.

Results. From τ_3 alone, oxygenation and composition are structurally non-identifiable because both enter a single summed decay rate. Ignoring composition produced apparent- pO_2 errors of -178, -439 and -860 mmHg at 2%, 5% and 10% lipid for a tissue at 20 mmHg. Adding I_3 restored identifiability in principle, but the pO_2 CRLB remained 330-760 mmHg at 10^5 ortho-positronium counts. A 10 mmHg target required litres of pooled signal even in the best-yield scenario and was not achievable at 4 mm voxel scale.

Conclusion. Positronium-lifetime hypoxia imaging is limited by an oxygenation-versus-composition identifiability problem. Lifetime-only maps should not be interpreted as oxygenation maps without composition control and a second observable.

Keywords: positronium lifetime imaging; PET/CT; hypoxia; oxygenation; identifiability; Cramér-Rao bound

Introduction

Ortho-positronium (o-Ps), the triplet bound state formed in a fraction of positron-electron annihilations, localises in nanometre-scale free-volume regions and annihilates in matter predominantly through pick-off processes. Its lifetime is therefore sensitive to tissue microstructure and composition. Molecular oxygen provides an additional paramagnetic quenching channel through spin conversion, making the o-Ps lifetime an attractive candidate biomarker for oxygenation and tumour hypoxia [1, 2]. This is clinically relevant because hypoxic tumour regions are more resistant to radiotherapy and systemic therapy, while established nuclear-medicine hypoxia tracers have practical limitations.

The technical basis for in-vivo positronium lifetime imaging (PLI) has advanced rapidly. Dedicated multi-photon systems and long-axial-field-of-view PET/CT scanners now support in-vivo positronium measurements and regional lifetime estimates [3-6]. These developments make the oxygenation question timely. However, the o-Ps lifetime is not oxygen-specific. Across soft tissues, composition-driven lifetime variation is on the order of hundreds of picoseconds, whereas the oxygen-specific term implied by aqueous quenching calibrations is on the order of picoseconds across physiological pO_2 differences [1, 7]. Calibrations performed in water or aqueous samples control composition by design and therefore do not test the in-vivo confound.

The positronium literature recognises this qualitatively: o-Ps lifetime reflects several phenomena, including oxygen, free radicals and free-volume structure [7]. What has been missing is a quantitative statement of whether oxygenation and composition are identifiable as separate variables, how large the apparent hypoxia bias can be, and how many counts would be needed to overcome the degeneracy with a second observable. Here we formulate the problem as an identifiability and CRLB analysis for an emerging nuclear-medicine biomarker. We quantify the bias produced by a lifetime-only, water-calibrated interpretation; test the structural rank of the lifetime-only inverse problem; evaluate whether adding o-Ps intensity I_3 or a related $3\gamma/2\gamma$ observable can break the degeneracy; and translate the resulting precision limits into clinically interpretable count and volume requirements.

Prior 3γ oxygenation experiments and recent proposals for joint lifetime and $3\gamma/2\gamma$ readouts motivate the use of a second observable [8-10]. A related single-subject cardiac PLI re-analysis reported an in-vivo lifetime contrast that was not explained by blood oxygenation, illustrating why the general identifiability problem matters in human data [11].

Materials and methods

Forward model

We modelled the o-Ps decay rate as a sum of independent channels:

$$\lambda_3 = 1/\tau_3 = \lambda_{\text{pickoff}}(f) + \kappa(f)\alpha(f)pO_2,$$

where f is the lipid/free-volume-rich fraction of a two-compartment tissue, $\lambda_{\text{pickoff}}(f)$ is the composition-dependent baseline rate, κ is the oxygen quenching constant and α is oxygen solubility. The baseline lifetime $\tau_0(f)$ interpolated between water and lipid endpoints, and the lipid:water oxygen-solubility ratio was anchored to published oil/lipid measurements [12, 13]. The o-Ps intensity was modelled as

$$I_3 = I_{3,w} + a_f f + a_p pO_2,$$

where $a_f = \partial I_3 / \partial f$ and $a_p = \partial I_3 / \partial pO_2$. The two load-bearing unknowns were made explicit: κ_{lipid} , swept over 0.1-1.0 times the water value, and the composition slope $\partial I_3 / \partial f$ (Table 1). The baseline scenario assumed the pessimistic case $\partial I_3 / \partial pO_2 \approx 0$, so that oxygen affects the lifetime rate but not o-Ps formation probability.

Apparent-pO₂ bias

We defined a naive reader as an estimator that assumes pure-water calibration ($f = 0$) and inverts an observed lifetime to pO₂. The resulting apparent-pO₂ error was computed over true pO₂ and lipid fraction. This bias quantifies the oxygenation error that would arise when a composition-dependent lifetime is interpreted as oxygen-specific.

Identifiability and CRLB analysis

For $\theta = (pO_2, f)$, Fisher information matrices were computed from τ_3 alone and from the two-observable vector (τ_3, I_3) . Structural non-identifiability was defined by rank deficiency of the Fisher information matrix; practical identifiability was assessed by the CRLB and parameter correlation. From τ_3 alone, both parameters enter through the scalar rate λ_3 , so the score vectors are collinear and the Fisher information has rank 1. With (τ_3, I_3) , the problem becomes identifiable only if the Jacobian of (λ_3, I_3) with respect to (pO_2, f) is non-collinear.

Poisson spectrum and nuisance profiling

To avoid treating τ_3 and I_3 as independent clean measurements, we built a binned Poisson lifetime-spectrum model. The expected spectrum contained a fast exponentially modified Gaussian component, an o-Ps component with slope $\tau_3(\theta)$ and amplitude $I_3(\theta)$, and a large flat accidental background. The per-bin Fisher information was

$$F_{ab} = \sum_k m_k^{-1} \partial_a m_k \partial_b m_k,$$

where m_k is the expected count in bin k . This includes the τ_3 - I_3 estimation covariance. We also profiled spectral nuisance parameters by extending the Fisher information matrix to include background level, fast-component lifetime, prompt offset and instrument resolution, then marginalising to (pO_2, f) . As an internal estimator check, Poisson spectra were simulated under the same forward model and fitted by maximum likelihood to verify attainability of the fixed-nuisance CRLB (Online Resource 1).

Counts and volume model

For the count-to-volume translation, we used a scenario anchor from a public ^{44}Sc PLI phantom preprint: approximately 3.6×10^4 o-Ps counts in a 5.57 mL region [14]. We scaled o-Ps counts by region-of-interest volume and by scenario-level effective isotope-yield factors representing prompt branching and source-to-background conditions. The scenarios were ^{124}I (best yield), $^{68}\text{Ga}/^{44}\text{Sc}$ (reference) and ^{82}Rb (lower yield). For each tissue state, we computed the region volume required to reach $\sigma(pO_2) \leq 10$ mmHg and classified a virtual tumour at 4 mm voxel scale as viable, marginal or not achievable.

All analyses were implemented in Python 3.13 using NumPy, SciPy and Matplotlib. Code, constants and generated results are available in the public repository listed in Data Availability. Use of AI-assisted tools is reported in Statements and Declarations.

Results

Lifetime-only interpretation produces large apparent oxygenation errors

The composition-dependent lifetime shift dominated the oxygen-specific lifetime term. For a tissue truly at $pO_2 = 20$ mmHg, a water-calibrated lifetime read produced apparent- pO_2 errors of -178 mmHg at 2% lipid, -439 mmHg at 5% lipid and -860 mmHg at 10% lipid (Fig. 1). These values are physiologically implausible, indicating that the uncorrected lifetime read is composition-dominated rather than oxygen-specific.

The oxygen sensitivity was also composition-dependent and uncertain because κ_{lipid} has not been measured in relevant lipid/protein environments. The model reproduced the aqueous sensitivity of approximately 0.09 ps/mmHg and predicted a net range of approximately 0.09-0.90 ps/mmHg in lipid-rich material across the κ_{lipid} sweep (Fig. 2). Thus, lipid affects both the baseline lifetime and the oxygen sensitivity.

τ_3 alone is structurally non-identifiable

From the lifetime alone, pO_2 and composition were not separately identifiable. Both parameters enter only through the single summed decay rate λ_3 , so the

two-parameter Fisher information matrix has rank 1 and the individual CRLBs diverge. This is an algebraic non-identifiability, not a shortage of counts.

Adding I_3 restored identifiability in principle, because composition moves both τ_3 and I_3 while, in the pessimistic baseline model, oxygen moves only the rate. The formal condition is a non-collinear Jacobian for (λ_3, I_3) with respect to (pO_2, f) (Fig. 3). If $\partial I_3 / \partial f$ approaches zero, the system again becomes degenerate. A non-zero $\partial I_3 / \partial pO_2$ can help if it supplies an additional non-collinear direction, but it is not sufficient by itself if it remains aligned with the lifetime-rate response.

The second-observable rescue is statistically weak

At 10^5 o-Ps counts and a representative hypoxic fatty state ($pO_2 = 20$ mmHg, $f = 0.10$), the pO_2 CRLB remained large across noise models: 332 mmHg for the fixed-nuisance Poisson spectrum, 570 mmHg for an idealised two-observable model using the field’s empirical lifetime precision, 665 mmHg after profiling background and fast-component nuisances, and 758 mmHg after profiling all tested nuisances (Table 2). The spectral model also exposed substantial τ_3 - I_3 covariance, with parameter correlation $\rho = 0.70$ in the optimistic Poisson model compared with $\rho = 0.05$ in the idealised two-observable model.

The count scaling implied approximately $3\text{--}5 \times 10^8$ o-Ps events for a 10 mmHg target in the profiled models, depending on the noise model (Fig. 4). The spectrum calculations are expressed in total in-window events, with $N_3 = I_3(1 - \text{background})N_{\text{total}}$; at the representative state used here, 10^5 o-Ps counts correspond to approximately 1.1×10^6 in-window events. Parametric Monte Carlo confirmed that the fixed-nuisance bound is statistically attainable under the assumed spectrum model: empirical standard deviations matched the CRLB at both tested count levels, and the pO_2 pull distribution had mean $+0.05$ and standard deviation 1.00 (Online Resource 1). This validates estimator behaviour under the specified model; it does not validate the unmeasured biological constants.

Voxel-scale pO_2 separation is not achievable under current yield assumptions

When translated to region-of-interest volumes, clinically useful separation required litre-scale pooling. At $pO_2 = 10$ mmHg and 5% lipid, the required pooled volumes were approximately 20 L for the best-yield ^{124}I scenario, 78 L for ^{68}Ga and 130 L for ^{82}Rb (Fig. 5). At 60% lipid, where oxygen sensitivity is higher, the best-yield scenario still required approximately 5 L. At a native 4 mm clinical voxel (0.064 mL), pO_2 separation was not achievable anywhere in the virtual tumour phantom (Fig. 6).

Discussion

This analysis identifies a fundamental limitation for positronium-lifetime hypoxia imaging. Oxygen quenching of o-Ps is real, but a lifetime map is not an oxygenation map unless composition is controlled. From τ_3 alone, oxygenation and composition collapse into a single measured decay rate, making the inverse problem structurally non-identifiable. This remains true even with arbitrarily many counts. The practical consequence is large apparent-pO₂ bias when composition is ignored.

The analysis is also constructive. It shows that a second observable, such as o-Ps intensity I_3 or a related $3\gamma/2\gamma$ ratio, is required to break the degeneracy. This requirement is analogous to adding an independent contrast dimension to separate two physical sources of signal, as in established multi-parameter imaging models [15]. In the pessimistic baseline model, the key separability handle is the composition dependence of I_3 . Therefore, PLI oxygenation studies should report the lifetime together with I_3 or $3\gamma/2\gamma$, and should treat co-registered CT/MRI-derived composition as a correction variable rather than a nuisance to ignore.

However, identifiability in principle does not imply clinical feasibility. The CRLBs show that current count yields are far from sufficient for voxel-level pO₂ separation. The relevant design target is not merely estimating a single lifetime, for which photon-limited precision scaling is well established [16], but separating oxygenation from composition in a two-parameter model. Under the yield scenarios used here, that target requires regional pooling at volumes incompatible with focal tumour hypoxia imaging. This does not preclude regional or ex-vivo studies, nor future systems with substantially higher effective o-Ps yields, but it sets a quantitative bar for claims of voxel-scale hypoxia imaging.

The decisive experiment is a controlled PALS matrix over O₂ by medium: water, lipid emulsion and protein/serum. Such an experiment should measure both τ_3 and I_3 under independently measured dissolved oxygen. It would determine κ_{lipid} , protein-environment quenching, $\partial I_3/\partial f$ and $\partial I_3/\partial pO_2$, converting the present conditional model into a measured one. This is a modest bench experiment compared with a full clinical imaging study and should precede strong clinical claims about PLI hypoxia.

This study has limitations. The forward model is intentionally transparent and two-compartment; it does not include richer Tao-Eldrup free-volume physics or chemical effects such as inhibition and radical scavenging. The I_3 values are illustrative placeholders, and the composition slope is assumed non-zero in the baseline scenario. The lipid quenching constant is unmeasured in relevant biological environments and was therefore swept. The count model is anchored to one public preprint yield and uses scenario-level isotope factors rather than scanner-specific measured yields. Nuisance profiling provides a more conservative CRLB, but the spectrum model remains idealised. Finally, the Monte Carlo validates statistical attainability under the assumed spectrum model, not the

biological forward model. A PALS measurement or re-analysis of real data with I_3 extraction remains the key empirical anchor.

In summary, positronium-lifetime hypoxia imaging is limited by an oxygenation-versus-composition identifiability problem. Lifetime-only PLI contrasts should not be interpreted as hypoxia without composition control. A second observable can make the problem identifiable, but current count yields imply that clinically useful voxel-scale separation is not achievable without major gains in effective o-Ps statistics or strong external composition constraints.

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Statements and Declarations

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Author contributions. E.D.Z. conceived the study, implemented the model, performed the analysis, generated the figures and wrote the manuscript.

Data availability. No clinical or animal datasets were analysed. The datasets generated during the current study, including code, constants, intermediate outputs and figures, are available at <https://github.com/Jefemaestro33/oxygenation-versus-tissue-composition-in-positronium-lifetime-pet-identifiability-limit>.

Code availability. The analysis code is available at <https://github.com/Jefemaestro33/oxygenation-versus-tissue-composition-in-positronium-lifetime-pet-identifiability-limit> under the repository licence terms.

Ethics approval. Not applicable. This is a modelling and methods study using no human participants, human data, animals or animal data.

Consent to participate. Not applicable.

Consent to publish. Not applicable.

Use of AI tools. AI-assisted programming and editorial tools were used to help cross-check code, draft text and identify consistency issues under the author's direction. The author reviewed all outputs and takes responsibility for the final manuscript, analysis and interpretation.

Tables

Table 1 Forward-model constants and load-bearing unknowns

Quantity	Value	Status	Source
τ_0 water	1.815 ns	measured	[1]
τ_0 adipose	2.5-2.7 ns	measured	[7]
$\kappa(\text{O}_2)$ water	0.0204 μmol^{-1} μs^{-1} L	measured	[1, 2]
O_2 solubility lipid:water	approximately $5\times$	measured	[12, 13]
Quenching mechanism	spin conversion	measured	[2]
κ_{lipid}	unmeasured; swept 0.1-1.0 times water	unknown	proposed experiment
$\partial I_3 / \partial f$	unmeasured; assumed non-zero in baseline scenario	unknown	proposed experiment

Table 2 Cramér-Rao bound on $p\text{O}_2$ at $p\text{O}_2 = 20$ mmHg, $f = 0.10$ and 10^5 o- Ps counts

Noise model	$\sigma(p\text{O}_2)$ (mmHg)	ρ
Poisson spectrum, nuisances fixed	332	+0.70
Idealised two-observable model	570	+0.05
Poisson spectrum, background and fast-component nuisances profiled	665	–
Poisson spectrum, all tested nuisances profiled	758	–

Figure legends

Fig. 1 Apparent $p\text{O}_2$ error produced by a water-calibrated lifetime interpretation when tissue composition is ignored. The contour marks the region where absolute error is 10 mmHg. Even low lipid fractions drive errors of hundreds of mmHg

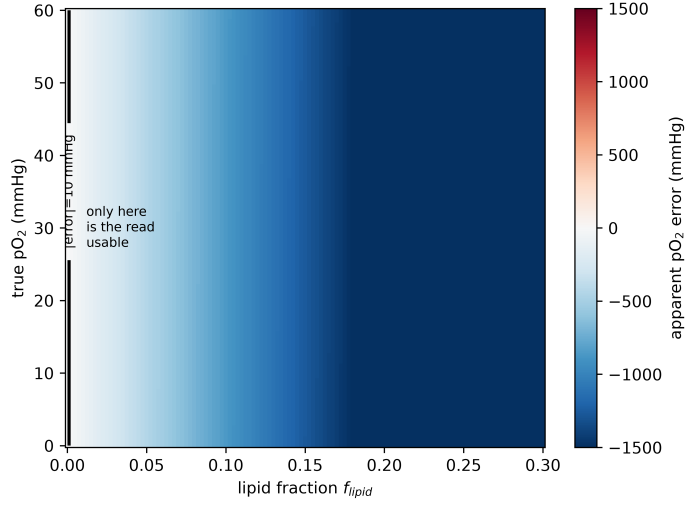


Fig. 2 Oxygen sensitivity of the o-Ps lifetime as a function of lipid fraction. The band represents the sweep over the unmeasured lipid quenching constant κ_{lipid} from 0.1 to 1.0 times the water value

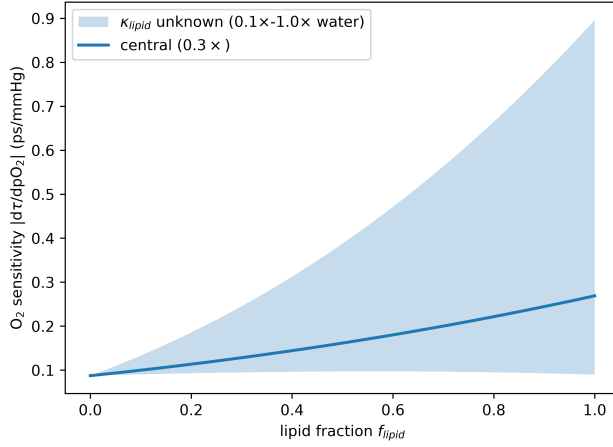


Fig. 3 Separability geometry of (τ_3, I_3) . Iso- τ_3 contours show the near-degenerate lifetime direction, while iso- I_3 contours provide the composition handle. The shallow crossing reflects identifiability in principle but weak statistical separation

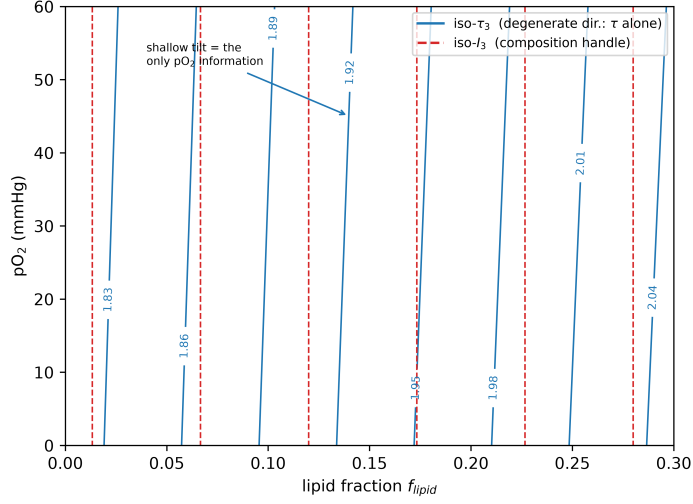


Fig. 4 CRLB $\sigma(pO_2)$ versus total in-window events for the full-spectrum (τ_3, I_3) fit. The lifetime-only case is structurally non-identifiable. A 10 mmHg target requires order- 10^9 in-window events, corresponding to order- 10^8 o-Ps counts after the $N_3 = I_3(1 - \text{background})N_{\text{total}}$ conversion

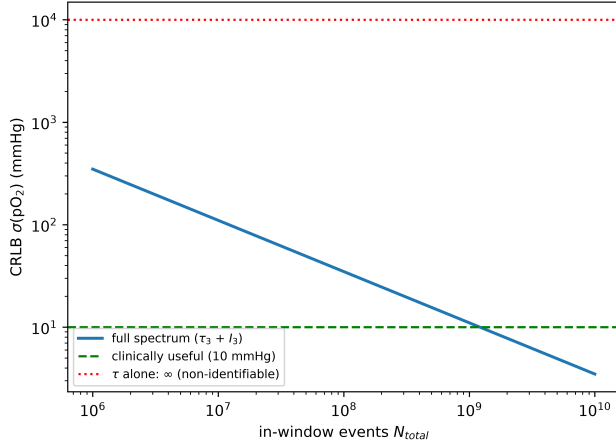


Fig. 5 Region-of-interest volume required to separate pO_2 from composition with $\sigma(pO_2) \leq 10$ mmHg for three effective isotope-yield scenarios. Clinical separation requires litre-scale pooling under current assumptions

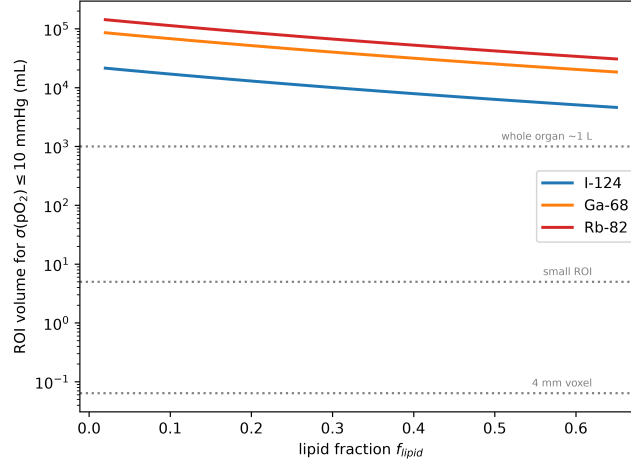


Fig. 6 Virtual tumour simulation. The left panel shows true pO_2 ; the right panel shows voxel-level separability at 4 mm resolution under the profiled CRLB model. pO_2 separation is not achievable anywhere at native voxel scale



Supplementary information

Online Resource 1 Parametric Monte Carlo validation of the fixed-nuisance Poisson CRLB under the assumed spectrum model. Pulls of $\hat{p}O_2$ are close to $N(0,1)$, showing that the maximum-likelihood estimator reaches the fixed-nuisance CRLB when the forward model is correctly specified.